



Memorandum

To: Tony Radich, US DOE on Detail to USDA
From: Jay Golden
Date: June 2, 2018
Ref: **Bioindicators Project**
Cc: Rob Handfield

Tony, I wanted to provide to you this update. I am hopeful we can connect for a telephone conference and I am also planning to come up to DC for a couple of days if you would have time to meet in person. Overall, there is great progress for both the written report and the web portal database.

Contract Update

Before I provide an update on the technical aspects, I did want to give you an overview of the contractual aspects. Our contract runs through September 29, 2018. We will have a draft of the 2017 Report for you to review by August 29, 2018. This will provide us a month to obtain your comments and edits and for us to make all required changes to the document. The updated website will be done in September as well.

According to our accounting department, we have billed for \$11,971 as of April 30th and fully expect to have all funds used for this project. Prior to his retirement, Harry Baumes indicated that funding had been secured to update the report and website for 2018 however, no further communication has been provided. We are very eager and excited to have the potential opportunity to continue to build on this indicators project for USDA.

Technical Update

BIOINDICATORS OF THE U.S. BIOBASED ECONOMY: UPDATE

The report is divided into four sections: (1) database, (2) report, (3) future work, and (4) elements of discussion.

DATABASE

All sections of the database (i.e., energy, agriculture, bioproducts) have been modified as follows:

- We have added 2016 and 2017, if available.
- We have updated the existing data.
- To simplify the comparison between bioindicators, we have unified the units.
- We have included the conversion factor we used for each bioindicator.
- We have updated the sources used and added the retrieved date.

In addition to these general updates, new data has been added to the database. In particular we have added 24 new energy bioindicators, 75 new agriculture bioindicators and 4 new bioproducts indicators (see tables 1 to 3 for more details).

Table 1. New energy bioindicators “added” to the database.

Bioindicator	Years
Number of ethanol plants operational	2011-2017
Number of ethanol plants under construction	2013-2017
Production volume of cellulosic feedstock ethanol	2010-2017
Total volume of biodiesel consumed for non-transportation	2010-2017
Number of wood pellets plants in the United States	2013-2017
Number of proposed wood pellets plants in the United States	2013-2017
Number of states which have a wood pellet production facility	2013-2017
Number of existing wood pellets plants under construction	2013-2017
Consumption of wood for electricity generation total	2010-2017
Consumption of wood for electricity generation electric sector	2010-2017
Consumption of wood for electricity generation industrial	2010-2017
Consumption of wood for electricity generation and useful thermal output	2010-2017
Total jobs created in the wood pellet industry	2010-2016
Number of municipal solid waste plants in the United States	2010-2016
Number of municipal solid waste plants proposed in the United States	2010-2016
Number of municipal solid waste plants that were put on standby in the United States	2010-2016
Number of municipal solid waste plants that where closed-shut down in the United States	2010-2016
Number of municipal solid waste plants in the United States	2010-2016
Electricity net generation by waste-to-energy in the electric power sector	2010-2017
Electricity net generation by waste-to-energy in the industrial power sector	2010-2017
Electricity net generation by waste-to-energy in the commercial power sector	2010-2017
Consumption of waste for electricity generation	2010-2017
Consumption of waste for electricity generation and useful thermal output	2010-2017
Number anaerobic digesters in livestock farms used for cogeneration	2010-2017

Table 2. New agriculture bioindicators “added” to the database.

Bioindicator	Years
Number of farms in the United States	2008-2017
Acreage of agriculture in the United States	2008-2017
Average farm size	2008-2017
Employment in agriculture in the United States	2008-2017
Total United States farmland	2008-2017
Volume of corn grown	2008-2017
Corn processed into high fructose corn syrup	2008-2017
Corn processed into glucose and dextrose	2008-2017
Corn processed into starch	2008-2017
Corn processed into fuel alcohol	2008-2017
Corn processed into alcoholic beverages	2008-2017

Corn used as cereal	2008-2017
Corn saved for seed	2008-2017
Total volume of soybeans grown in the United States	2008-2017
Total volume of soy consumed for crush	2008-2017
Total volume of soy consumed for seed, feed and residual	2008-2017
Total volume of cotton imported	2008-2017
Direct value of all soy grown in the United States	2008-2017
Corn processed into starch	2008-2017
Yield of corn grown in the United States	2008-2017
Vegetable oil produced in the United States	2008-2017
Total volume of cotton grown in the United States	2008-2017
Total volume of peanuts grown in the United States	2008-2017
Total volume of edible tallow produced in the United States	2008-2017
Total volume of soy consumption	2008-2017
Total volume of cotton consumption	2008-2017
Total volume of sunflowers consumption	2008-2017
Total volume of canola consumption	2008-2017
Total volume of peanuts consumption	2008-2017
Total volume of corn consumption	2008-2017
Total volume of lard consumption	2008-2017
Total volume of edible tallow consumption	2008-2017
Soy imports	2008-2017
Cotton imports	2008-2017
Sunflowers imports	2008-2017
Canola imports	2008-2017
Peanuts imports	2008-2017
Corn imports	2008-2017
Lard imports	2008-2017
Edible tallow imports	2008-2017
Soy exports	2008-2017
Cotton exports	2008-2017
Sunflowers exports	2008-2017
Canola exports	2008-2017
Peanuts exports	2008-2017
Corn exports	2008-2017
Lard exports	2008-2017
Edible tallow exports	2008-2017
Volume of soybeans crushed	2008-2017
Soybean oil going to biodiesel	2008-2017
Percent soybean oil used for biodiesel	2008-2017
Volume of cottonseed crushed	2008-2017
Volume of sunflowers crushed	2008-2017

Volume of canola crushed	2008-2017
Canola oil going to biodiesel	2008-2017
Percent canola oil used for biodiesel	2008-2017
Volume of peanuts crushed	2008-2017
Corn oil going to biodiesel	2008-2017
Percent corn oil used for biodiesel	2008-2017
Value of soybeans used for biodiesel	2008-2017
Value of canola used for biodiesel	2008-2017
Value of corn used for biodiesel	2008-2017
Planted acreage of corn in the United States	2008-2017
Planted acreage of corn for ethanol production in the United States	2008-2017
Corn beginning stocks	2008-2017
Corn ending stocks	2008-2017
Corn consumption for feed	2008-2017
Corn consumption for food	2008-2017
Fertilizer used in corn	2008-2017
Acreage of soybeans planted in the United States	2008-2017
Acreage of soybeans harvested in the United States	2008-2017
Yield of soy grown in the United States	2008-2017
Fertilizer used in soy	2008-2017
Soy beginning stocks	2008-2017
Soy ending stocks	2008-2017

Table 3. New bioproducts bioindicators “added” to the database.

Bioindicator	Years
Biotech revenues in the United States	2008-2012
Global production capacity of biodegradable bioplastics	2008-2017
Value of chemical shipments	2008-2016
Value of chemical exports	2008-2016

REPORT

We have updated all figures in the report and added new ones considering the new data collected. In addition, we have generated 16 maps of the energy section and 39 maps of the agriculture section to be added in the new report (see tables 4 and 5 for more detail). While we have generated 55 new GIS maps to date, it is anticipated we will produce additional maps over the next two months.

Table 4. Maps of the energy section.

Bioindicator	Year
Energy consumption total	2015
Energy consumption commercial sector	2015
Energy consumption industrial sector	2015
Energy consumption residential sector	2015
Energy consumption transportation sector	2015
Average gas and diesel retail price by region	2018
Average biodiesel B20 retail price by region	2018
Average compressed natural gas and gasoline retail price by region	2018
Average ethanol and gasoline retail price by region	2018
Ethanol biorefinery production capacity	2017
Ethylene crackers location	2018
Wood pellets plant's output by tons	2017
Wood pellets manufacturing facility's status	2017
Biodiesel production capacity	2017
Biodiesel plant location	2017
Waste to energy plant location	2018

Table 5. Maps of the agriculture section.

Bioindicator	Year
Net cash income of farm operations	2012
Number of farm operations	2012
Percentage of change of gross cash income of farms	2010-2015
Percentage of change of fuel and oil expenses of farms	2010-2015
Percentage of change of direct government payments of farms	2010-2015
Percentage of change of livestock and poultry expenses of farms	2010-2016
Acreage of agriculture	2012
Acres of barley planted	2017
Acres of canola planted	2017
Acres of corn planted	2017
Acres of cotton planted	2017
Acres of peanuts planted	2017
Acres of sorghum planted	2017
Acres of soybeans planted	2017

Acres of sunflowers planted	2017
Acres of wheat planted	2017
Production of barley	2012 and 2017
Production of canola	2017
Production of corn for grain	2017
Production of cotton	2012
Production of peanuts	2017
Production of sorghum for grain	2012
Production of soybean	2017
Production of sunflower	2017
Production of wheat	2012 and 2017
Barley sales	2012
Corn sales	2012
Sorghum sales	2012
Wheat sales	2012
Top 10 the United States corn export markets	2017
Top 10 the United States cotton export markets	2017
Top 10 the United States soybeans export markets	2017
Top 10 the United States wheat export markets	2017
Top 10 sources of the United States fruits imports (fresh or frozen)	2017
Top 10 sources of the United States fruits imports (prepared or preserved)	2017
Top 10 sources of the United States vegetables imports (fresh or frozen)	2017
Top 10 sources of the United States vegetables imports (prepared or preserved)	2017

ON-GOING WORK

1. Updating the policy section
2. Produce additional new maps in the report
3. Create a second database with all geographical information used to generate the maps
4. Update the website with new visualizations

ELEMENTS OF DISCUSSION

We need your input regarding the following elements in the report's update:

1. Last year's 2016 report did comparisons to the previously published 2011 Bioindicators report at the request of Harry. Do you want to have this year's report to also compare data with the 2011 report? Or do we want to compare it with the 2016 report or both?
2. There are some bioindicators in the 2016 database (excel file) that our team could not verify (we highlight these values in yellow in the database), do we want to delete them or keep them?